Municipal Waste Management

Content

Municipal Waste Management

- Waste Stream and components
- Methods of Waste Disposal
- Recycling and Reusing Waste: Problems & Benefits
- Reducing Waste Stream: Necessity & Urgency
- Hazardous & Toxic Waste
- Sustainable Waste Management
- Circular Economy Concept

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Waste and the Waste Stream

Garbology: the science of trash

What can the stuff we throw away every day tell us about our society – and ourselves?
Professor William Rathje, UA

- Consumption pattern
- perception on waste from cradle –to-grave
- Behavior

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Waste Management



Letstute

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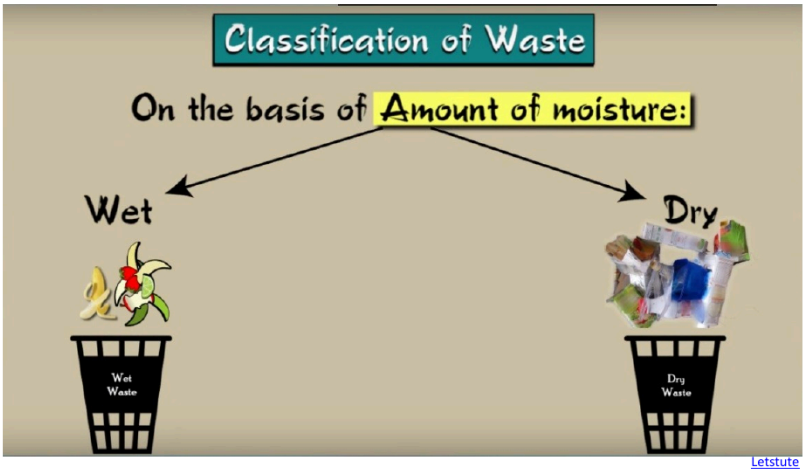
Waste and the Waste Stream



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Waste and the Waste Stream



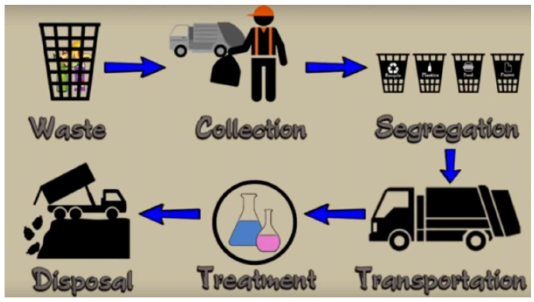
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Municipal Waste Management

Municipal waste: (also referred to as 'municipal solid waste') waste from households, as well as other waste which, because of its nature or composition, is similar to waste from households. (EPRS)

Municipal Solid Waste – Residential and commercial solid waste generated within a community. (UNEP)



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Waste Management

Need to segregate



Need to transport



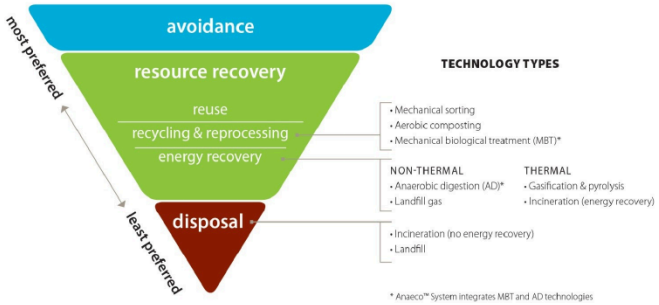
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Waste Management

Two principles apply across all waste streams:

The **'waste hierarchy'** sets a priority order among five waste treatment methods:
1) prevention, then 2) preparing for re-use,
3) recycling and 4) (energy) recovery, with
5) disposal as the least preferred option.

European Model



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Waste Management

Waste Scoreboard - **EUROPEAN COMMISSION, 2012**

Criterion / EU MS	1.1 Decoupling of waste from consumption	1.2 Waste Prevention Programme	1.3 Amount of municipal waste recycled	1.4 Amount of municipal waste recovered (energy recovery)	1.5 Amount of municipal waste disposed	1.6 Development of municipal waste recycling	2.1 Existence of ban/restrictions for the disposal of municipal waste into landfills	2.2 Total typical charge for the disposal of municipal waste in a landfill	2.3 Existence of pay-as-you-throw (PAYT) systems for municipal waste	3.1 Access to waste collection services	3.2 Available treatment capacity for municipal waste	3.3 Forecast of municipal waste generation and treatment capacity in the Waste Management Plan	3.4 Existence and quality of projection of municipal waste generation and treatment	3.5 Compliance of landfills for non-hazardous waste	4.1 Fulfilment of the targets related to biodegradable municipal waste going to landfills	4.2 Rate of biodegradable municipal waste going to landfills	5.1 Number of infringement procedures – WFD and Landfill Directives	5.2 Number of court cases – WFD and Landfill Directives	Over all score
AT	0	2	2 D	2 D	2 D	2	2	1	2	2	2	2	2	2	2	2	2	2	39
NL	0	2	2 D	2 D	2 D	2	2	2	1	2	2	2	2	2	2	2	2	2	39
DK	0	0	2 D	2 D	2 D	2	2	2	1	2	2	2	2	2	2	2	2	2	37
DE	1	0	2 D	1 D	2 D	2	2	2	2	2	2	2	1	2	2	2	2	2	36
SE	1	2	2 D	2 D	2 D	2	2	2	1	2	2	0	0	1	2	2	2	2	35
BE	1	2	2 D	2 D	2 D	2	2	2	1	2	2	0	0	2	2	2	1	1	34
LU	0	0	2 D	2 D	2 D	2	2	2	1	2	2	0	0	2	2	2	2	2	33
UK	1	2	2 D	1 D	2 D	2	0	1	1	2	2	2	1	1	2	1	2	2	32
FI	1	2	1 D	2 D	1 D	0	1	1	2	2	2	2	1	1	2	2	2	2	31
FR	1	2	1 D	2 D	2 D	1	1	1	1	2	2	2	1	1	2	2	1	1	31
SI	2	0	2 D	1 D	1 D	2	1	2	2	0	2	0	0	0	2	1	1	2	25
ES	2	0	1 D	1 D	1 D	1	0	1	1	2	2	0	0	1	2	1	1	1	21
PT	0	2	0 D	2 D	1 D	1	0	0	0	2	2	2	2	2	0	0	1	1	21
HU	1	0	1 D	1 D	1 D	2	0	0	1	0	0	0	0	2	2	1	2	2	19
IE	0	2	1 D	1 D	1 D	1	1	2	1	0	2	2	0	2	0	0	0	0	19
CZ	2	0	0 D	1 D	1 D	2	0	1	1	2	2	0	0	1	0	0	1	2	18
PL	1	2	1 D	0 D	1 D	2	1	1	1	0	2	0	0	1	0	0	1	2	18
EE	2	0	1 D	0 D	0 D	0	1	1	1	0	2	0	1	2	2	1	1	1	17
SK	2	0	0 D	1 D	0 D	1	1	0	1	2	2	0	0	1	2	1	1	1	17
IT	0	0	1 D	1 D	1 D	0	1	2	1	2	0	0	0	0	2	1	0	0	15
LV	2	0	0 D	0 D	0 D	1	1	1	0	0	2	0	1	2	0	0	2	2	14
CY	0	0	1 D	0 D	0 D	2	0	0	0	2	0	0	1	0	0	0	2	2	11
RO	2	0	0 D	0 D	0 D	1	0	0	0	0	2	0	1	0	0	1	2	2	11
LT	2	0	0 D	0 D	0 D	1	1	0	1	0	0	0	0	0	0	0	2	2	9
MT	0	0	0 D	0 D	0 D	2	0	0	0	2	0	0	0	2	0	0	1	2	9
BG	2	0	0 D	0 D	0 D	0	0	0	0	0	0	0	0	0	2	1	1	2	8
GR	1	0	0 D	0 D	0 D	0	0	0	0	2	0	0	0	0	0	0	0	0	3

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Waste Management

Waste Scoreboard - **EUROPEAN COMMISSION, 2012**

Criteria in full

1.1 Level of decoupling of municipal waste generation from household final consumption expenditure
1.2 Existence of own waste prevention program (WPP) or equivalent existence in Waste Management Plan or other (environmental) program
1.3. Amount of municipal waste recycled (material recycling and other forms of recycling including composting)
1.4. Amount of municipal waste recovered (energy recovery)
1.5. Amount of municipal waste disposed (deposit onto or into land and incinerated without energy recovery)
1.6. Development of municipal waste recycling (material recycling and other forms of recycling including composting)
2.1. Existence of nationwide ban/restrictions for the disposal of municipal waste into landfills
2.2. Total typical charge for the disposal of municipal waste in a landfill
2.3. Existence of pay-as-you-throw (PAYT) systems for municipal waste
3.1. Access to waste collection services for municipal waste
3.2. Available treatment capacity for municipal waste in line with the EU waste legislation (including disposal and incineration)
3.3. Forecast of municipal waste generation and treatment capacity in the Waste Management Plan
3.4. Existence and quality of projection of municipal waste generation and treatment in the Waste Management Plan
3.5. Compliance of landfills for non-hazardous waste with Landfill Directive
4.1 Fulfilment of the targets of the Landfill Directive related to biodegradable municipal waste going to landfill
4.2. Rate of biodegradable municipal waste going to landfills
5.1. Number of infringement procedures – Waste Framework Directive and Landfill Directive
5.2. Number of court cases – Waste Framework Directive and Landfill Directive

Sustainable Waste Management

Sustainability means taking into account in our work and in our daily lives all the resources upon the planet: humans, animals, plants, water, air, land space [etc] and managing them to grow and develop the resources that today exist, for the future.

Using material resources efficiently to cut down on the amount of waste produced. And, where waste is generated, dealing with it in a way that actively contributes to the economic, social and environmental goals of sustainable development.

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Sustainable Waste Management

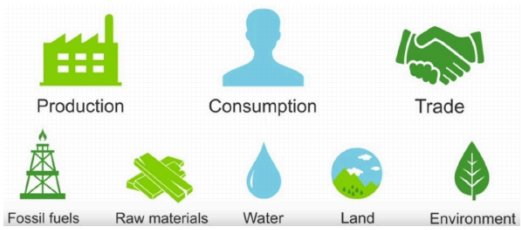
Instead of Approaches….

…it could be an **economic theory!!!**

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Sustainable Waste Management

"**Circular Economy** – can be defined as a system in which production, consumption and trade are designed to minimize the net extraction of key resources; such as fossil fuel, raw materials, water, land and the environment in pollutants along with diminishing negative environmental and health impacts" EU



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Sustainable Waste Management

Today's linear 'take, make, dispose' economic model relies on large quantities of cheap, easily accessible materials and energy, and is a model that is reaching its physical limits.

A **circular economy** is restorative and regenerative by design, and aims to keep products, components, and materials at their highest utility and value at all times.

Principle (s)

Principle 1: Preserve and enhance natural capital
...by controlling finite stocks and balancing renewable resource flows.

Principle 2: Optimize resource yields
...by circulating products, components, and materials at the highest utility at all times in both technical and biological cycles.

Principle 3: Foster system effectiveness
...by revealing and designing out negative externalities.

Ellen Macarthur Foundation
https://www.ellenmacarthurfoundation.org/circular-economy

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